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الجامعة العربية المفتوحة - مصر
Arab Open University
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الامتحانات
Arab Republic of Egypt

Faculty of Business Studies

FIN241

Microfinance

Mid-Term Exam VB

Spring 25-26
Date: 16 – April – 2026

Number of exam pages: 5 (including this cover page)	Time Allowed: 1.5 hours

Instructions:

1. Take FIVE (5) minutes to read all instructions carefully before attempting to write.
2. Mobile phones must be in "SWITCHED OFF" mode throughout the examination.
3. This is a CLOSED book examination. NO documents of any kind are allowed.
4. You MUST answer the questions using only the answer booklets provided
5. You MUST return the examination paper together with your answer booklets.
6. Do NOT write your answers in pencil except graphs/sketches
7. Students are NOT allowed to use any type of dictionary or mobile phones during the exam
8. Students are allowed to use non-programmable calculators only (if needed).
9. The penalty for an ACADEMIC MISCONDUCT in this examination is an F Grade.

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Part A: Solve the following problem (40 points)

Problem 1:

Given: The balance sheet and income statements of XYZ Microfinance.

BALANCE SHEET	Dec. 2021	Dec. 2020
ASSETS		
Cash	172,500	185,000
Investments	395,250	545,000
Total loan portfolio	4,100,000	3,600,000
(Loan loss reserve)	(335,250)	(285,000)
Total net loans outstanding	**3,764,750**	**3,315,000**
Fixed assets	162,500	138,000
(Accumulated depreciation)	(52,500)	(48,000)
Net fixed assets	**110,000**	**90,000**
TOTAL ASSETS	**4,442,500**	**4,135,000**
LIABILITIES		
Forced savings	575,000	585,000
Commercial loans	52,500	57,500
Subsidized loans	178,500	191,500
TOTAL LIABILITIES	**806,000**	**834,000**
EQUITY		
Donated equity cumulated	2,150,000	2,315,000
Donated equity current	0	285,000
Cumulated earnings/losses	361,500	(285,000)
Current earnings/losses	1,125,000	986,000
TOTAL EQUITY	**3,636,500**	**3,301,000**
TOTAL LIABILITIES + EQUITY	**4,442,500**	**4,135,000**

INCOME STATEMENT	Dec. 2021	Dec. 2020
OPERATING INCOME		
Interests and fees from loans	2,188,250	1,549,000
Income from investments	134,750	100,000
TOTAL OPERATING INCOME	**2,323,000**	**1,649,000**
OPERATING EXPENSES		
Interest and fee expenses	12,000	10,000

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Loan loss provision expense	185,000	50,000
Salaries	780,000	620,000
Other administrative expenses	195,000	225,000
Depreciation	26,000	8,000
TOTAL OPERATING EXPENSES	**1,198,000**	**913,000**
NET OPERATING PROFIT / LOSS	**1,125,000**	**736,000**
NON OPERATING INCOME		
Cash donations	0	250,000
Other non-operating income	0	0
TOTAL NON OPERATING INCOME	**0**	**250,000**
Non-operating expenses	0	0
TOTAL PROFIT / LOSS	**1,125,000**	**986,000**

You are also given the following expected rates:

inflation	**5.00%**
commercial loans	**22.00%**
subsidized loans	**22.00%**
savings deposits	**6.00%**
Capitalization rate	**15.00%**

Required:

Estimate the annual percentage rate (APR) of XYZ MFI.

Part B: Choose 2 out of 3 questions (60 points)

Question 1:

Non-bank financial institutions are amongst the institutions providing Microfinance services. Discuss the main traits of these institutions and their role in providing microfinance initiatives. Support your answer with relevant examples. **(30 marks)**

Question 2:

Informal or community-based financial service providers can be divided into two broad categories: indigenous and facilitated. Discuss money lenders, Pawn brokers and Deposit collectors or money guards. Support your answer with relevant examples. **(30 marks)**

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Question 3:
How can the government promote savings and ensure consumer protection, according to what you have seen in the microfinance course? **(30 marks)**

****************End of questions****************

Formula Sheet

- APR = (AE + LL + CF + K –II) / (1-LL)
- AE = (salaries + rent + depreciation + other administrative expenses) / Average outstanding portfolio
- LL = ((Loan loss reserve year(n-1) + Loan loss provision year(n))- Loan loss reserve year(n) / average outstanding portfolio
- Average equity for cost of funds calculation = Average equity – average net fixed assets
- Forced savings = Average forced savings x rate on saving services
- subsidized loan= Average subsidized loans x rate on commercial loans
- Commercial loan = Average commercial loans x rate on commercial loans
- Equity= Average equity for cost of funds calculation x inflation rate
- CF = (Forced savings + subsidized loan + Commercial loans + Equity) / Average outstanding portfolio. note: the numbers in the numerator are used after applying the market rates
- II = Income from other than loan portfolio / Average outstanding loans
- Arrears Rate = Total loan installments past due / Gross portfolio outstanding
- Portfolio at Risk (PAR) = Total outstanding loan balance on delinquent loans / Gross portfolio outstanding
- Loan Loss rate = total amount written off (in the period) / Average outstanding portfolio (in the period)
- INFLATION ADJUSTMENT = (Average equity – Average net fixed assets) x Inflation rate
- COST OF FUNDS ADJUSTMENT = Average subsidized funds x (commercial interest rate – interest rate paid) OR *(Average subsidized funds x commercial rate) - interest rate paid*
- OSS = operating income / total operating expenses
- FSS = Adjusted operating income / adjusted operating expenses

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Ratio	Formula
Portfolio to assets	Gross loan portfolio/assets
Operating efficiency ratio	Operating expense/revenues
Cost per active client	Operating expense/average number of active clients
Borrowers per loan officer	Number of active borrowers/ number of loan officers
Active clients per staff member	Number of active clients/total number of personnel
Portfolio yield	Cash received from interest, fees, and commissions on loan portfolio/ average gross loan portfolio
Return on assets (ROA)	Net income after taxes and before donations/average assets
Return on equity (ROE)	Net income after taxes and before donations/average equity
Nonperforming loans as of 30 days past due (NPL30)	(Nonperforming loans> 30 days + value of renegotiated loans)/gross loan portfolio
Renegotiated loans ratio	Value of renegotiated loans/gross loan portfolio
Write-off ratio	Value of loans written off/average gross loan portfolio
Debt to equity (leverage)	Liabilities/equity
Equity to assets ratio	Total equity/total assets
Liquid ratio (quick ratio or acid-test ratio)	(Cash + trade investments)/ (demand deposits + short-term time deposits +short-term borrowings + interest payable on funding liabilities +accounts payable + other short-term liabilities)
Liquid assets to total assets	Cash/total assets
Loans to deposits ratio	Gross loan portfolio/deposits

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